Thermodynamic consistency and closure limits of radiatively cooled X-point collapse

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Version 1.0.0; 29 October 2026

Abstract

We examine the thermodynamic reduction in Chowdhry and Loureiro's model of current-sheet formation under radiative cooling, using its journal and arXiv editions. The displayed pressure formula does not solve the stated energy equation. Direct integration gives a Bernoulli solution whose initial cooling rate is nonzero, whereas the published bremsstrahlung form has zero initial cooling rate for every finite spatial integration coefficient. Correcting this formula generally destroys the exact parabolic-pressure affine closure. On the common positive-pressure branch, the corrected scalar evolution preserves pressure ordering; in the published final ordinary differential equations, outflow deceleration while the inflow scale is shorter than the outflow scale requires negative central total pressure. Separately, a cumulative cooling integral cannot be replaced by its vanishing endpoint tail. At fixed finite coefficients and finite positive outflow scale, that integral makes the printed cooling-to-magnetic force ratio tend to zero, contradicting the proposed strict cooling-dominated endpoint balance under those hypotheses. We retain the algebra of the modified steady Sweet-Parker branch and state its parameter-domain and stability limitations. Direct scalar integrations and separate high-precision checks support these bounded conclusions. They do not reconstruct the original numerical trajectories or rule out physical radiative contraction in a different model.

1. Target and scope

Radiative cooling can change the pressure and compression of reconnecting layers. Chowdhry and Loureiro [1,2] propose an affine X-point-collapse model and a separate modification of a radiatively cooled Sweet-Parker layer. Their abstract connects strong cooling to accelerated inflow collapse and arrested or reversed outflow elongation. The pressure reduction is therefore central to the proposed dynamical explanation, rather than a peripheral normalization choice.

We examine the 26-page journal version of record, *Journal of Plasma Physics* **91**(6), E148 (2025), DOI 10.1017/S0022377825100949 [1], and the supplied 24-page arXiv:2508.13081v1 [2]. The supplied preprint is byte-identical to the official versioned PDF. Both editions have been read completely. The final article corrects the preprint's missing equality in Eq. (2.14), updates the flow-rate symbols in Eq. (3.1), and adds cooling-law, Newtonian-MHD and relativistic-applicability qualifications. Those improvements are credited. The pressure construction, principal collapse balances and steady-layer expressions addressed here remain in the final edition. Locators below refer to [1] unless stated otherwise.

The main question is internal consistency: do the displayed reductions satisfy the energy equation and their own positivity and asymptotic assumptions? We derive the pressure evolution first, then examine its spatial implications. Separate checks preserve the steady-layer algebra instead of treating every conclusion as invalid. No original numerical analysis program or complete set of execution inputs was obtained from the inspected source archive. The original curves are not reproduced here.

2. The pressure initial-value problem

Let $J(t) = \xi(t)\eta(t) > 0$ be the affine Jacobian, with the source's normalization $J(0) = 1$, and $\rho_L = \rho_0/J$. We use $\bar{C}$ for the positive coefficient already defined by the source's Eq. (2.22), including its conversion factors. Equation (2.23) is

$$\frac{d}{dt}\ln\left(\frac{P_L}{\rho_L^{\gamma}}\right) = -\bar{C}\rho_L^{a-b}P_L^{b-1}. \tag{1}$$

The derivative follows a fixed fluid label. Define

$$Y = P_L J^{\gamma}, \quad q = b(1-\gamma)+\gamma-a, \quad \kappa = \bar{C}\rho_0^{a-b}, \quad I(t) = \int_0^t J(t')^q\,dt'.$$

Then Eq. (1) gives $\dot{Y} = -\kappa J^q Y^b$, with $Y(0) = P_0$. On the positive branch its unique solution is

$$P_L = J^{-\gamma}\left[P_0^{1-b} - (1-b)\kappa I(t)\right]^{1/(1-b)}, \qquad b \neq 1, \tag{2}$$

and

$$P_L = J^{-\gamma}P_0\exp[-\kappa I(t)], \qquad b = 1. \tag{3}$$

For $b < 1$, Eq. (2) is used only while its bracket is positive. Continuing a squared bremsstrahlung expression beyond its zero would reverse the sign of the square root used in the derivation and cease to solve the same positive-pressure equation. These formulas require no special value of the initial pressure and no choice of an additional spatial integration function.

In contrast, after choosing $C_2 = P_0$, the source's Eqs. (2.24)-(2.27) and (A5)-(A6) have the functional form

$$Y_{\rm pub} = P_0\left[1 - \widetilde{C}I(t)^{1/(1-b)}\right]. \tag{4}$$

This expression puts the initial pressure outside the nonlinear power. It is not Eq. (2). The source also asserts that an independent spatial C_2 remains unconstrained after specifying the initial condition. A first-order scalar initial-value problem on $P_L > 0$ has no such freedom. In Appendix A, the equality between $d\ln F/dt$ and $F^{-b}\dot{F}$ in Eq. (A4) already changes the left-hand side; adding homogeneous and particular solutions does not repair a nonlinear equation by superposition.

The bremsstrahlung case gives a normalization-independent counterexample. With $b = 1/2$, Eq. (2) requires

$$Y = \left(\sqrt{P_0} - \frac{\kappa I}{2}\right)^2, \qquad \dot{Y}(0) = -\kappa\sqrt{P_0} \neq 0. \tag{5}$$

Every finite coefficient multiplying I^2 in Eq. (4) instead gives $\dot{Y}_{\rm pub}(0) = 0$ for regular $J(0) > 0$. Changing that finite coefficient cannot recover the missing derivative. For the elementary prescribed history $J = 1$, $P_0 = \kappa = 1$, the correct pressure is $(1 - t/2)^2$ until $t = 2$; a printed-form example with $\widetilde{C} = 1/4$ is $1 - t^2/4$. At $t = 1$ these give $1/4$ and $3/4$. The numerical coefficient in this illustration is declared, not inferred from an original plotted run.

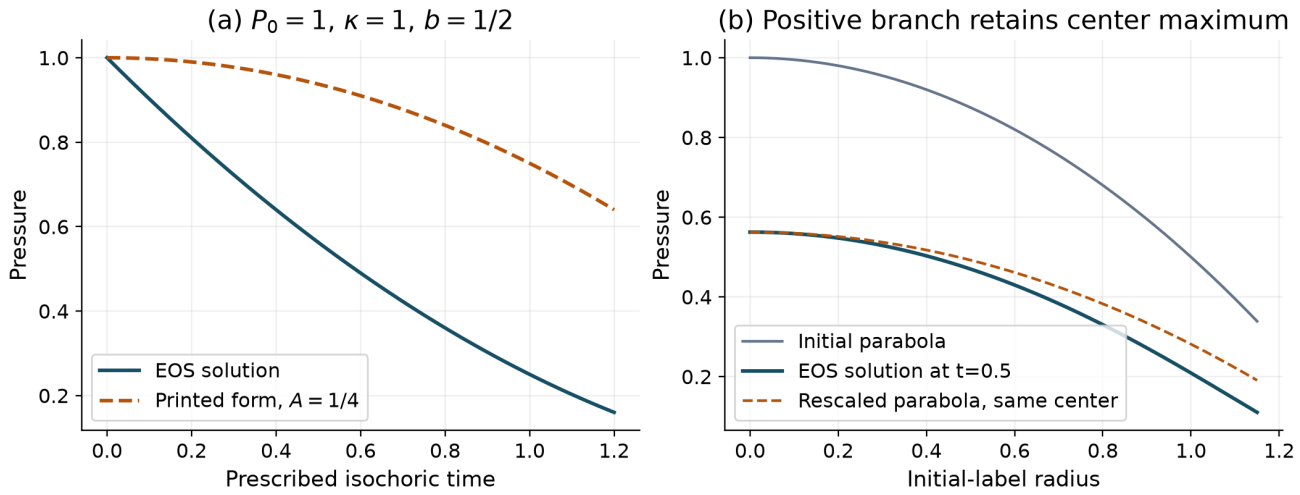


Figure 1. Left: the isochoric bremsstrahlung solution has a nonzero initial cooling slope; the printed functional form does not. Right: at $t = 0.5$, an initially parabolic profile remains centrally maximal but is not a common rescaling of that parabola. The dashed comparison is matched at the center.

The discrepancy is not universal across all exponents. At $b = 0$, a specifically calibrated additive coefficient can reproduce Eq. (2); it is not arbitrary. At $b = 1$, the exponential solution must be treated separately. These controls help distinguish the nonlinear integration error from a blanket rejection of every cooling law.

3. Spatial closure and the pressure sign

The source chooses spatially uniform initial density and $P_0 = P_{0,X}[1 - (x_0^2 + y_0^2)/2]$, then assumes a spatially uniform $J(t)$. For fixed common J, differentiating the repaired solution with respect to its positive initial value gives

$$\frac{\partial P_L}{\partial P_0} = J^{-\gamma} \left(\frac{Y}{P_0}\right)^b > 0. \tag{6}$$

Thus pressure ordering is preserved while both compared labels remain on the positive branch. The initially higher central pressure cannot fall below a lower surrounding pressure in this scalar, uniform-density affine construction. Different fractional cooling rates do not by themselves reverse the ordering. Once a label reaches the boundary of the assumed cooling law, this comparison makes no assertion about additional physics beyond it.

For $b = 1/2$, the factor in Eq. (6) is $J^{-\gamma}[1 - \kappa I/(2\sqrt{P_0})]$. It depends on position through P_0. Multiplying the initial parabolic gradient by this factor generally produces a non-affine force. Consequently, inserting the correct pressure into momentum does not simply yield the source's two scale-factor equations with a repaired time-only cooling factor. Special cases such as $b = 0$, $b = 1$, uniform pressure or zero cooling must be separated. A local Taylor expansion about the origin could define an approximate closure, but that would be a new approximation requiring its own validation, not the claimed exact global reduction.

There is also a direct positivity restriction inside the source's *printed* final Eq. (2.31). For bremsstrahlung, write $\theta = 1 - \widetilde{C} I^2$ and $D = \xi^{-2} - \eta^{-2}$. Its outflow acceleration becomes

$$\ddot{\eta} = \xi \left[D + \frac{P_{0,X}\theta}{(\xi\eta)^\gamma} \right]. \tag{7}$$

If $0 < \xi < \eta$, $P_{0,X} > 0$ and $\theta \geq 0$, every term on the right is nonnegative and $D > 0$. Hence $\ddot{\eta} > 0$. Deceleration on this branch requires $\theta < 0$, which is negative central *total thermal pressure* in Eq. (2.26). Negative pressure is necessary here, not sufficient to outweigh the magnetic term. The pressure is not defined as a freely adjustable perturbation; adding a constant background changes an absolute-pressure nonlinear cooling law and requires solving that law again.

This restriction matters for the interpretation in Section 3.3, which invokes stronger nonuniform compression and a pressure maximum turning into a minimum. The stated affine Jacobian and uniform initial density instead give spatially uniform density at each time. Neither that proposed density mechanism nor a positive-pressure central minimum follows from the equations supplied. This does not prove that a nonuniform-density plasma, a non-affine flow or a more complete radiative-MHD model cannot contract along its outflow direction.

4. Cumulative cooling and endpoint balances

A separate issue remains even if the published ODE is examined formally. For $a = 2$, $b = 1/2$, $\gamma = 5/3$, Eq. (2.29) defines $I = \int_0^t (\xi\eta)^{-2/3} dt'$. Suppose $\tau = t_c - t$, $\eta \to \eta_c > 0$ and $\xi \sim A\tau^\alpha$, with no earlier singularity. For the proposed $\alpha = 2/3$ and $\alpha = 1$, the endpoint is integrable:

$$I_c - I(t) \sim \frac{(A\eta_c)^{-2/3}}{1 - 2\alpha/3}\, \tau^{1-2\alpha/3}, \qquad 0 < I_c < \infty. \tag{8}$$

It is the tail $I_c - I$, rather than the integral from the initial time, that vanishes. The scaling substitution in Section 3.2 and Appendix B drops this positive cumulative constant. It is leading order, not a negligible integration term. In particular, the ratio of the printed radiative-force magnitude to its inverse-square magnetic term is

$$\mathcal{D} = P_{0,X} \widetilde{C} I(t)^2 \eta^{-\gamma} \xi^{2-\gamma} \sim \text{const.}\, \tau^{\alpha/3} \longrightarrow 0. \tag{9}$$

This limit uses fixed finite coefficients and a finite positive η_c. It contradicts a strict endpoint balance dominated by that cooling term under those same hypotheses, including the proposed intermediate equality in Eq. (3.13). A large coefficient can delay the crossover greatly. The argument does not reject a finite-window effective exponent near one, a joint infinite-cooling limit, or a branch where η also vanishes. It also does not determine the physical endpoint of the repaired, non-affine thermodynamic problem. Substituting exactly $\alpha = 1$ into a second-derivative leading balance additionally requires care because $\alpha(\alpha - 1) = 0$.

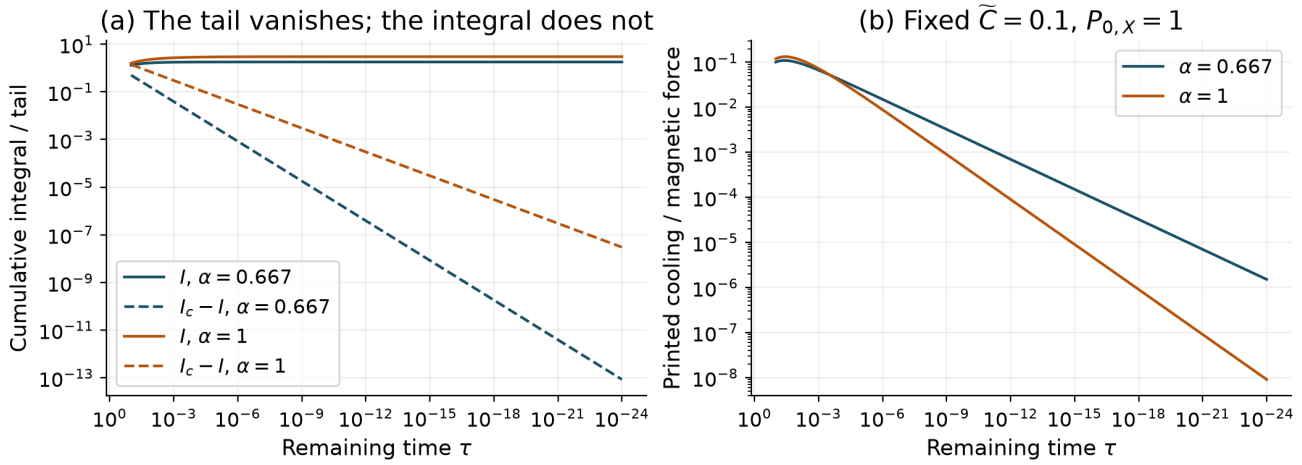


Figure 2. For $t_c = 1$, $\eta = 1$, $\xi = \tau^\alpha$, the cumulative limit is $I_c = 1/(1 - 2\alpha/3)$: 1.8 or 3. The force ratio uses fixed $P_{0,X} = 1$ and $\widetilde{C} = 0.1$. The printed pressure factor remains positive in both examples. The small remaining-time values illustrate an analytic limit; they are not measured physical collapse times.

The leading inverse-square scalar equation also fixes a coefficient: $\ddot{\xi} = -\eta_c/\xi^2$ requires $A^3 = 9\eta_c/2$ for $\xi = A\tau^{2/3}$. Appendix B's Eq. (B14) instead prints $9\eta_c/4$, giving half the required acceleration. The missing factor is visible between its kinetic-energy first integral and quadrature. This is a discrepancy in the target's displayed calculation, not an allegation about the classical literature it cites.

Retaining a generic finite net thermal coefficient has a further consequence. Define $H_c = P_{0,X}\eta_c^{1-\gamma}(\widetilde{C}I_c^2 - 1)$. The frozen leading scalar equation is $\ddot{\xi} = -\eta_c\xi^{-2} - H_c\xi^{-5/3} + \cdots$. When $H_c \neq 0$, its first thermal correction is

$$\xi = A\tau^{2/3} + D_1\tau^{8/9} + \cdots, \qquad D_1 = \frac{81}{44}H_cA^{-5/3}. \tag{10}$$

This follows by linearizing the inverse-square force as well as expanding the second derivative. A free scalar-energy correction of order $\tau^{4/3}$ is recovered in the zero-thermal-force control; it is not generically the first thermal correction when $H_c \neq 0$. These are local consistency expansions of reduced equations, not a global existence theorem for the source initial data.

5. Normalization and the steady branch

Some local printed discrepancies can be identified without using the pressure solution. Using the approximate right-hand side printed in Eq. (2.33) and the outflow time in Eq. (2.34), with $D_0 = \bar{C}\rho_0^{a-b}P_{0,X}^{b-1}$, their ratio is $R = D_0/\Gamma_{y,0}$. Equation (2.35) instead multiplies by $\Gamma_{y,0}$. A unit normalized initial rate masks the difference. The literal pressure-over-radiated-power definition would give $D_0/[(\gamma - 1)\Gamma_{y,0}]$; either clock retains division by the outflow rate. Equation (2.36) also reverses $1 - b$ to $b - 1$ relative to Eq. (2.27); bremsstrahlung's squared exponent masks that sign. Neither discrepancy establishes which values an unavailable plotting implementation used. These clock-convention factors are distinct from the energy-equation failure.

The separate steady model deserves a positive control. Its continuity, Ohm's-law, power and cooling/advection balances give Eqs. (4.7)-(4.16), including $A \approx 1 + \beta$, consistently within their displayed approximations. Here $A = \rho_X/\rho_{\rm in}$ denotes compression, rather than the collapse amplitude, and $\beta = \beta_{\rm in}$. We retain these balances. With $c = (\gamma - 1)/\gamma^{3/2}$, the bremsstrahlung branch is

$$\lambda \equiv \frac{L}{L_{\rm sys}} = \frac{c}{\mathcal{R}\beta(1+\beta)}, \qquad F \equiv \frac{v_{\rm in}/V_A}{S_{\rm sys}^{-1/2}} = \left[\frac{\mathcal{R}\beta(1+\beta)^2}{c}\right]^{1/2}. \tag{11}$$

Here $\mathcal{R}$ is the source's system-scale cooling parameter, not the collapse-time R. The comparison is with the classical incompressible Sweet-Parker scaling [4]. The source already neglects order-one outflow factors; Eq. (11) uses its displayed conventions, not an exact coefficient calibration.

Nevertheless, $\mathcal{R} \gg 1$ alone is not uniform in upstream beta. A layer fitting the system requires $\mathcal{R} > c/[\beta(1+\beta)]$. Enhancement requires $\mathcal{R} > c/[\beta(1+\beta)^2]$. At fixed $S_{\rm sys}$, increasing cooling indefinitely also leaves the thin, slow-inflow branch:

$$\frac{\delta}{L} = \left[\frac{\mathcal{R}\beta}{cS_{\rm sys}}\right]^{1/2}, \qquad \frac{v_{\rm in}}{V_A} = \left[\frac{\mathcal{R}\beta(1+\beta)^2}{cS_{\rm sys}}\right]^{1/2}. \tag{12}$$

For $\gamma = 5/3$ and $S_{\rm sys} = 10^8$, three examples distinguish a retained branch from extrapolations outside it:

β	$\mathcal{R}$	$L/L_{\rm sys}$	F	Interpretation
1	100	0.001549	35.93	Fits, thin, sub-Alfvenic
10^{-6}	100	3098	0.01797	Layer exceeds system
1	10^{10}	1.549×10^{-11}	3.593×10^5	Not thin or slow inflow

The low-beta example is not an admissible counterexample to a branch explicitly restricted to $L < L_{\rm sys}$. It shows why that restriction must accompany a conclusion based only on large $\mathcal{R}$. At fixed positive beta and suitable Lundquist number, the source's shortened-layer enhancement remains possible.

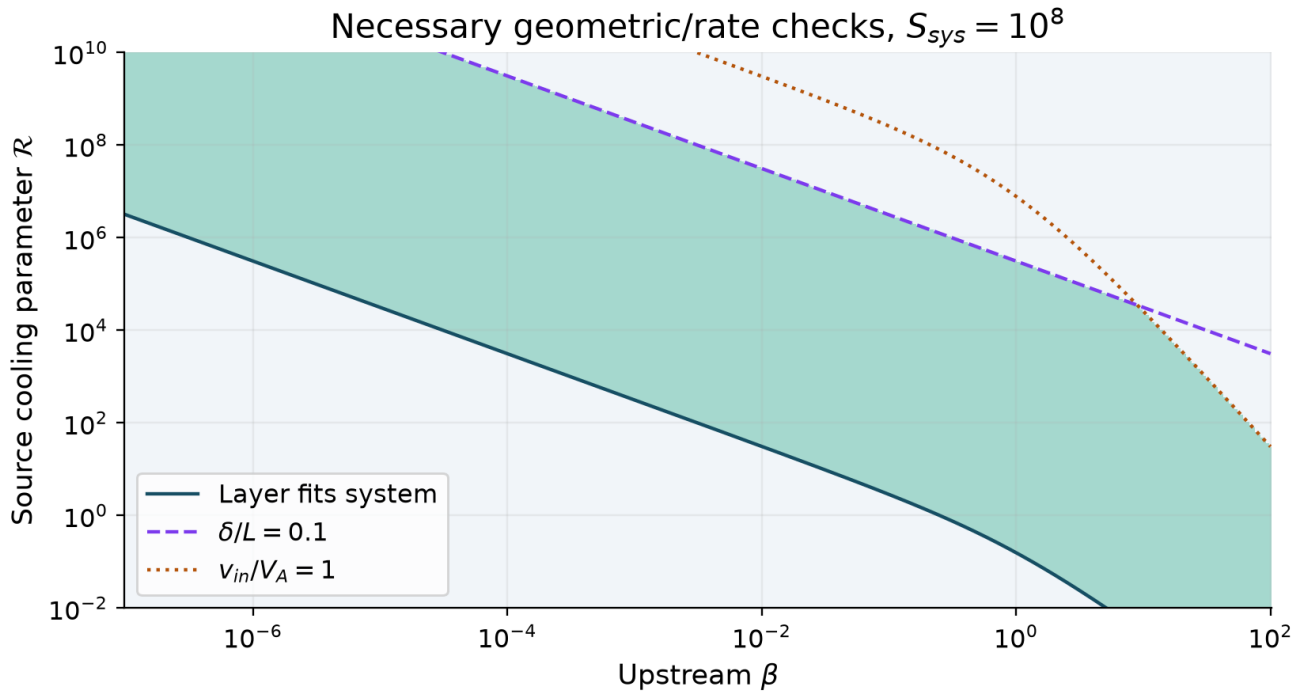


Figure 3. Shading satisfies $L < L_{\text{sys}}$, $\delta/L < 0.1$ and $v_{\text{in}} < V_A$ under the displayed conventions. The 0.1 threshold is an illustrative thinness choice. No numerical threshold for $\mathcal{R} \gg 1$, optical thinness, collisional validity or thermal stability is imposed; the shading alone does not certify them.

Finally, an algebraic steady solution does not establish thermal stability or evolutionary accessibility. Uzdensky and McKinney [3, Section 4.1] explicitly distinguish these questions for their model and derive a bremsstrahlung instability condition. Allowing length to vary changes the possible branch, but the target supplies no dynamical length-response law or perturbation spectrum proving its stability. We do not transfer the earlier fixed-closure instability criterion unchanged to this new branch, nor infer stability from the mere existence of Eqs. (4.14)-(4.16).

6. Verification, limitations and conclusion

Three original figures and deterministic scalar examples accompany independent verifiers [5]. The pressure verifier integrates the original scalar pressure equation directly with DOP853, rather than integrating its transformed solution. Independent 65-digit quadrature then evaluates Eqs. (2)-(3). Eighty parameter/history reference controls, within 318 scalar integrations, agree to a maximum relative pressure difference 9.76×10^{-16} in the recorded environment. Twenty groups of eleven ordered initial pressures retain ordering. Spatial differences of separately integrated pressures agree with high-precision derivatives to a maximum absolute difference 3.36×10^{-10}; the bremsstrahlung force coefficient varies across labels while the $b = 0, 1$ controls remain spatially constant. All 4,096 seeded nonnegative-pressure outflow states, including one zero-pressure control, satisfy Eq. (7)'s strict acceleration sign.

A separate 150-digit Decimal calculation compares 48- and 96-point scalar quadratures, checks cumulative limits down to $\tau = 2^{-320}$, and verifies the coefficient and reduced-equation subleading controls. Another independent calculator checks nine cooling-clock examples, six rational real-power sign cases, 24 parameter cases checking the steady-state identities, and four branch-domain cases. The verifiers do not import the plotting producer. Agreement is numerical and algebraic evidence, not physical accuracy, independent human review or bitwise reproducibility across arbitrary environments.

The source authors' five figures, unprovided numerical execution parameters and original trajectories

are not re-created. The pressure examples impose volume histories solely to test the energy equation. No full MHD/PIC calculation, repaired global affine evolution, physical collapse time or empirical replacement for the source's critical cooling threshold is claimed. Non-affine compression, density variation, heating, conduction, resistivity and breakdown of a cooling law can change the problem. A quoted numerical bremsstrahlung coefficient remains under-specified in temperature/composition conventions and is not promoted as a verified error. Minor presentation discrepancies are recorded separately from the main findings.

The supported correction is substantive but bounded: the displayed pressure solution fails its governing equation, and restoring it removes the proposed exact global closure in the generic bremsstrahlung case. Positive-pressure restrictions and the retained cumulative integral further limit the stated flow-reversal and endpoint interpretations. The separate steady-branch algebra survives, accompanied by its necessary domain and unanswered stability questions.

Code and data availability. Original code, numerical outputs, figure sources, source-edition hashes and portable independent verifiers are archived as software version 1.0.0 [5], `https://doi.org/10.5281/zenodo.23780790` , direct record `https://zenodo.org/records/23780790` . The frozen computational commit is `8a4e64cc032cb0c57bb621b200903b0c0f0706c2` in `https://github.com/Teylersf/radiative-cooling-current-sheet-audit` . The anonymous archive download matches all 36 frozen files, and a clean extraction reproduced all four programs without changing those bytes. The DOI was independently verified as `findable` in its primary DataCite registry on 29 October 2026; a Crossref-only lookup does not test this registration. Source PDFs are linked through their official records and are not bundled as this author's work.

Funding and disclosures. Taylor Kalin self-funds this study. No outside funding or competing interests have been reported. AI tools assisted with source retrieval, derivation, code, adversarial checks and drafting. These checks are not independent human peer review. The source authors and institutions have not endorsed this work. Original code is MIT licensed; original audit text and figures are CC BY 4.0. Attributed third-party material retains its rights.

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